DSA0309 – NATURAL LANGUAGE PROCESSING

ASSIGNMENT – 04

Semantic Analysis and Semantic Systems Engineering

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GitHub Link: https://github.com/YOGAMADHA/DSA0309-NLP/tree/main/Assignment%204

PART I – SEMANTIC ANALYSIS THROUGH STRUCTURED REASONING

This part is a design-and-analysis assignment. The answers below show how a sentence can be converted into semantic units, structured representations and reasoned interpretations. Source code is not required for Part I.

Task 1 – Meaning Representation and Language Structure (8 Marks)

1.1 Identify Meaning Components (2 Marks)

Sentence: "Maya gave Ravi a book yesterday."

Component	Identification
Entities	Maya, Ravi and a book
Event	GIVE – a transfer of an object
Agent / giver	Maya
Recipient	Ravi
Theme / transferred object	A book
Temporal information	Yesterday

The linguistic form is a past-tense declarative sentence with the subject "Maya", verb "gave", indirect object "Ravi" and direct object "a book". Its meaning is an event in which Maya is the giver, Ravi receives the book, and the event happened yesterday. The grammatical positions are the surface form; the event and participant roles are the represented meaning.

1.2 Compare Meaning Representations (3 Marks)

(a) Conceptual / event structure

GIVE(e1) | AGENT(e1, Maya) | RECIPIENT(e1, Ravi) | THEME(e1, book) | TIME(e1, yesterday)

This representation identifies one giving event and assigns a role to each participant. It is compact and makes the event relationships explicit. It leaves details such as the book's identity, the exact date and the manner of transfer unspecified.

(b) Structured frame / slot representation

Frame slot	Value
Event type	Giving / transfer
Event identifier	e1
Agent	Maya

Recipient	Ravi
Theme	a book
Tense	Past
Time	Yesterday

The frame is easy for a computer system to store because every value has a named slot. The event structure emphasizes relations between participants, while the frame emphasizes a fixed set of fields. Neither representation identifies the exact book or the precise clock time.

1.3 Meaning Structure and Ambiguity (2 Marks)

Word order helps identify the grammatical subject and objects, but semantic roles explain what each participant does in the event. In the original sentence, Maya is the giver, Ravi is the recipient and the book is the theme.

Alternative sentence: “Yesterday, a book was given to Ravi by Maya.”

Its passive construction changes the word order, but its core representation remains GIVE(e1), AGENT(e1, Maya), RECIPIENT(e1, Ravi), THEME(e1, book), TIME(e1, yesterday). The semantic roles remain the same even though “a book” becomes the grammatical subject.

1.4 Design Defence (1 Mark)

The event representation is suitable for a computer because it stores meaning as labelled relations rather than depending only on sentence order. A program can query who performed the action, who received the object and when it happened. The frame also provides fixed fields for validation and storage. These structures can be compared, searched and used by later NLP components. They are therefore more useful for computation than a simple paraphrase.

Task 2 – First Order Predicate Calculus (8 Marks)

2.1 Define the Logical Vocabulary (2 Marks)

Symbol	**Type**	**Meaning / argument order**
Student(x)	Predicate	x is a student
Course(c)	Predicate	c is a course
Instructor(i)	Predicate	i is an instructor
Enrolled(x,c)	Predicate	student x is enrolled in course c
Teaches(i,c)	Predicate	instructor i teaches course c
Passed(x,c)	Predicate	student x passed course c
DrRao, NLP	Constants	Dr. Rao and the NLP course
x, c, i	Variables	A student, course and

		instructor

The first argument of Enrolled and Passed is the student; the second is the course. The first argument of Teaches is the instructor and the second is the course.

2.2 Translate Natural Language into Logic (3 Marks)

(i) Every student is enrolled in at least one course:

$\forall x\ [\text{Student}(x) \rightarrow \exists c\ (\text{Course}(c) \wedge \text{Enrolled}(x,c))]$

(ii) Some student passed every course taught by Dr. Rao:

$\exists x\ [\text{Student}(x) \wedge \forall c\ ((\text{Course}(c) \wedge \text{Teaches}(\text{DrRao},c)) \rightarrow \text{Passed}(x,c))]$

(iii) No instructor teaches every course:

$\forall i\ [\text{Instructor}(i) \rightarrow \exists c\ (\text{Course}(c) \wedge \neg\text{Teaches}(i,c))]$

The third formula states that each instructor has at least one course that they do not teach. It assumes that the course domain contains at least one course.

2.3 Interpret and Check the Formulae (2 Marks)

For formula (i), the universal quantifier $\forall x$ applies to every individual x. For each individual who is a student, the existential quantifier $\exists c$ requires at least one course in which that student is enrolled. The existential course may be different for different students.

Quantifier order changes meaning. Compare:

A: $\forall x\ (\text{Student}(x) \rightarrow \exists c\ \text{Enrolled}(x,c))$ – every student has at least one enrolled course.

B: $\exists c\ (\text{Course}(c) \wedge \forall x\ (\text{Student}(x) \rightarrow \text{Enrolled}(x,c)))$ – there is one course in which every student is enrolled.

Counterexample model: Let the students be Asha and Bala, and the courses be NLP and AI. Suppose Asha is enrolled only in NLP and Bala is enrolled only in AI. Formula A is true because each student has a course. Formula B is false because there is no single course shared by both students. Thus the two formulae are not logically equivalent.

2.4 Logical Consistency (1 Mark)

The pair cannot both be true if "NLP" refers to the same course and the domain is the same. "Every student passed NLP" means $\forall x\ [\text{Student}(x) \rightarrow \text{Passed}(x,\text{NLP})]$. "There is a student who did not pass NLP" means $\exists x\ [\text{Student}(x) \wedge \neg\text{Passed}(x,\text{NLP})]$. The second statement directly contradicts the first for that student. Therefore, the pair is inconsistent.

Task 3 – Linguistically Relevant Concepts and Syntax-Driven Semantic Analysis (9 Marks)

3.1 Identify Linguistically Relevant Concepts (2 Marks)

Sentence: "The experienced researcher saw the student with the telescope."

Concept	Example / role	Why it matters
Entities	researcher, student, telescope	Identifies participants and objects
Action	saw	Represents the seeing event
Modifier	experienced	Describes the researcher
Agent / experiencer	researcher	Person who performs the seeing
Theme / perceived entity	student	Person being seen
Prepositional phrase	with the telescope	May describe the student or the seeing event

3.2 Syntax-to-Semantics Mapping (3 Marks)

Attachment A – noun attachment: “with the telescope” modifies “the student”. The student has or is carrying the telescope.

Representation: SEE(e1) ∧ EXPERIENCED(researcher) ∧ AGENT(e1,researcher) ∧ THEME(e1,student) ∧ HAS(student,telescope).

Attachment B – verb attachment: “with the telescope” modifies “saw”. The researcher used the telescope as an instrument.

Representation: SEE(e1) ∧ EXPERIENCED(researcher) ∧ AGENT(e1,researcher) ∧ THEME(e1,student) ∧ INSTRUMENT(e1,telescope).

A syntax-driven procedure first parses the sentence into noun phrases, the verb phrase and the prepositional phrase. It then checks which constituent the prepositional phrase attaches to. If it attaches inside the object noun phrase, HAS(student,telescope) is produced. If it attaches to the verb phrase, INSTRUMENT(e1,telescope) is produced.

3.3 Semantic Composition (2 Marks)

1. Identify the nouns: researcher, student and telescope.
2. Combine “experienced” with “researcher” to form the noun phrase “experienced researcher”.
3. Combine the subject noun phrase with “saw” to create a seeing event.
4. Combine “the student” as the theme or perceived participant.
5. Attach “with the telescope” to either the student noun phrase or the seeing event.

The interpretations diverge at the attachment step. The word meanings remain the same, but the relation assigned to the telescope changes.

3.4 Failure Analysis (2 Marks)

Possible error	Effect	Validation check
Wrong prepositional	System says the student has a telescope when the researcher	Check attachment against the parse tree and nearby

attachment	actually used it.	verb/noun relations; retain both readings if evidence is weak.
Assigning roles from word order alone	A passive or reordered sentence may receive incorrect agent and theme roles.	Use grammatical dependencies and semantic-role constraints; verify that the assigned role is compatible with the predicate.

Task 4 – Lexemes, Senses and Word Sense Disambiguation (9 Marks)

4.1 Lexemes and Senses (2 Marks)

A lexeme is an abstract vocabulary unit that groups related word forms and meanings. A word form is the actual written or spoken form in a sentence. A sense is one particular meaning of a lexeme.

Context	**Word form / lexeme**	**Part of speech**	**Sense**
Nila deposited cash at the bank.	bank / BANK	Noun	Financial institution
They sat on the bank of the river.	bank / BANK	Noun	Land beside a river
The pilot banked the aircraft to the left.	banked / BANK	Verb, past tense	Tilt or turn an aircraft

The noun “bank” in A and B has different senses. In C, “banked” is a verb form of the lexeme BANK and describes an aircraft movement.

4.2 Design a WSD Method (3 Marks)

Input: a sentence containing the target word “bank”. Output: the selected sense label and the evidence used.

Candidate senses: FINANCIAL_INSTITUTION, RIVER_EDGE and AIRCRAFT_TURN. The system converts the sentence to lowercase, tokenizes it and examines a context window of up to five words on either side of the target. It checks for cue words:

Sense	**Context cues**
Financial institution	cash, deposit, money, account, loan, ATM
River edge	river, water, shore, fishing, sat
Aircraft turn	aircraft, pilot, plane, wing, left, turn

Decision rule: count the cue words for each candidate sense. Select the sense with the highest score if it has a positive score and is the unique highest candidate. If all scores are zero or there is

a tie, return UNCERTAIN rather than guessing. The output contains the target, sense, scores and evidence.

4.3 Test the Method (2 Marks)

Sentence	Evidence	Selected sense
Nila deposited cash at the bank.	deposited, cash	Financial institution
They sat on the bank of the river.	river, sat	River edge
The pilot banked the aircraft to the left.	pilot, aircraft, left	Aircraft turn
The bank approved my home loan.	approved, loan	Financial institution

A likely failure is “The bank was crowded.” The context contains no strong cue to distinguish a financial institution from another possible meaning. The method should return UNCERTAIN. A larger context or a clarification from the user would help.

4.4 Evaluation (2 Marks)

A small labelled test set can contain 30 sentences: 10 financial, 10 river-edge and 10 aircraft-turn examples. Two annotators should label each sentence using the same sense definitions; disagreements should be reviewed before testing. The system prediction is compared with the gold label.

Accuracy = (number of correct predictions / total test instances) × 100. For example, if 26 of 30 predictions are correct, accuracy = (26/30) × 100 = 86.67%. A balanced set gives each sense equal representation, but real text may contain far more financial uses than aircraft uses. High balanced accuracy therefore may not reflect performance on naturally imbalanced data. New domains and unfamiliar context can also reduce reliability.

Task 5 – Information Retrieval and Semantic Matching (8 Marks)

5.1 Query and Document Representation (2 Marks)

Query: “methods for finding the meaning of ambiguous words”. The central concepts are identifying a word’s intended sense, using surrounding context, and resolving ambiguity.

Represent the query as concept labels such as WORD, AMBIGUITY, CONTEXT, SENSE and DISAMBIGUATION. Represent each document using its important terms, concepts, topic and a short semantic description. This permits matching even when the query and document use different wording.

5.2 Lexical versus Semantic Retrieval (2 Marks)
Keyword retrieval compares the exact words in the query and document. It is simple and fast, but may miss a document that uses different wording. Semantic retrieval compares concepts and meaning, so it can find relevant material even when exact words differ.

Example of a document keyword matching may miss: “Contextual language models select the intended interpretation of polysemous terms from surrounding text.” It does not repeat the exact phrase “finding the meaning of ambiguous words”, but it describes word-sense disambiguation. Semantic retrieval can connect these concepts.

5.3 Ranking Design (2 Marks)
Use three signals: (1) lexical overlap, (2) concept similarity and (3) field relevance. Define:

FinalScore(d) = 0.30 × Lexical(d) + 0.50 × ConceptSimilarity(d) + 0.20 × FieldRelevance(d).

Each signal is normalized to a value from 0 to 1. Concept similarity receives the largest weight because the task is meaning-based retrieval. Field relevance gives additional weight to documents whose title or abstract directly discusses WSD. Sort by final score from highest to lowest. If scores tie, prefer the document with higher concept coverage, then use document ID as a stable final tie-breaker.

5.4 Evaluation and Error Analysis (2 Marks)
Precision = relevant retrieved documents / all retrieved documents. Recall = relevant retrieved documents / all relevant documents in the collection.

Example: There are 4 relevant documents in the collection. The system retrieves 5 documents, of which 3 are relevant. Precision = 3/5 = 0.60 (60%). Recall = 3/4 = 0.75 (75%).

A false positive may be a dictionary article that lists definitions but does not explain how context selects the intended sense. A false negative may be a useful article titled “Resolving polysemy in context” that uses few query words. Improvements include adding concept expansion, checking whether a document explains a method, and using human-labelled relevance examples.

Task 6 – Integrated Semantic Analysis Challenge (8 Marks)
Message: “I saw her duck near the bank, and it was closed.”

Stage 1 – Interpret the Problem (1 Mark)
The word “duck” may be a noun (a bird belonging to her) or a verb (an action of lowering her head/body). “Bank” may mean a financial institution or the land beside a river. “It” may refer to the bank or another nearby entity. The word “closed” strongly suggests an institution or facility, but the full context is still needed.

Stage 2 – Build Candidate Interpretations (2 Marks)
Interpretation A: The speaker saw a duck (bird) belonging to or associated with a woman near a financial bank. The bank was closed.

Representation: SEE(e1) ∧ AGENT(e1,speaker) ∧ THEME(e1,duck_bird) ∧ NEAR(duck_bird,financial_bank) ∧ CLOSED(financial_bank).

Interpretation B: The speaker saw the woman lower her head near a river bank, and the river bank/area was closed to access (for example, a restricted site).

Representation: DUCK(e2) ∧ AGENT(e2,her) ∧ LOCATED_NEAR(e2,river_bank) ∧ CLOSED(river_bank).

The second reading is less natural with "it was closed", but it is possible if the river-side area is a restricted location. The system should keep alternatives until context supports one.

Stage 3 – Syntax-Driven Analysis and WSD (2 Marks)
The parser checks whether "her" is a determiner/possessive before the noun "duck" or an object pronoun before the verb "duck". It then checks the grammatical role of "near the bank" and the antecedent of "it". The adjective "closed" is compatible with a bank branch or a restricted facility, making the financial-institution reading more likely. Useful extra context includes whether the speaker was discussing a bank visit, a river walk, or a bird.

Stage 4 – Information Retrieval Trace (2 Marks)
If the financial reading is selected, create the query concepts: BANK_BRANCH, OPENING_HOURS, CLOSURE_STATUS and CUSTOMER_SUPPORT. Candidate documents may include: (1) "Why a bank branch may be closed"; (2) "How to check branch opening hours"; (3) "Riverbank habitat and access restrictions"; (4) "Duck species and bird behaviour".

Rank	Document	Reason
1	Why a bank branch may be closed	Matches financial-bank sense and closed status
2	How to check branch opening hours	Matches branch availability and support intent
3	Riverbank habitat and access restrictions	Shares bank-related wording but wrong selected sense
4	Duck species and bird behaviour	Matches duck as a bird but not the closure support need

The system uses sense compatibility as a ranking signal. A document containing the word "bank" is not automatically relevant; it must match the selected financial sense and the closure issue.

Stage 5 – Challenge and Refine (1 Mark)
Misleading context: "We were watching birds beside the river when someone said, 'I saw her duck near the bank, and it was closed.'" A system that relies only on "closed" may incorrectly

select a financial bank. The refinement is to score the full discourse context and allow more than one candidate sense. If the confidence difference is small, the system should ask whether “bank” means a bank branch or a river bank.

PART II – SEMANTIC SYSTEMS ENGINEERING

Part II moves from analysis to a reproducible design, trace, test and evaluation. The Python project supplied separately is an optional rule-based demonstration. It supports the main task areas but does not claim to be a full trained NLP system.

Task 1 – Representing Meaning and Language Structure (8 Marks)

1.1 Semantic Structure Analysis (2 Marks)

Sentence: "Leena sent the report to Arun after the meeting."

Element	Value
Event	SEND
Agent	Leena
Theme	the report
Recipient	Arun
Temporal expression	after the meeting
Implicit information	The exact date/time and the report identity are not specified.

The surface form is a past-tense sentence with Leena as subject, sent as verb, report as object and to Arun as a recipient phrase. The semantic representation identifies the sending event and the roles independently of word order.

1.2 Build Meaning Representations (3 Marks)

(a) Semantic-role/event frame:

SEND(e1); AGENT(e1, Leena); THEME(e1, report); RECIPIENT(e1, Arun); TIME_RELATION(e1, AFTER(meeting)).

(b) Graph / slot representation:

Node / relation	Value
Event node	e1 : SEND
AGENT edge	e1 → Leena
THEME edge	e1 → report
RECIPIENT edge	e1 → Arun
TIME edge	e1 → AFTER(meeting)

The event frame makes participant roles explicit. The graph makes connections between nodes visible and can be extended with other events. Neither representation gives the meeting's exact time or the report's contents.

1.3 Meaning-Preserving Variation (2 Marks)
Alternative: "After the meeting, the report was sent to Arun by Leena."

Invariant relations: Leena remains the agent, the report remains the theme, Arun remains the recipient, and the sending occurs after the meeting. A meaning-changing variation would be "Leena sent Arun's report to Maya after the meeting," because the theme ownership and recipient are different.

1.4 Representation Check (1 Mark)
Validation rule: every SEND event must contain exactly one identified agent and one theme; when the sentence contains an explicit "to" recipient, the frame must also contain a recipient. If any required slot is empty or assigned to the wrong entity, flag the representation for review.

Task 2 – First Order Predicate Calculus: Formalise and Test (9 Marks)

2.1 Predicate Vocabulary (2 Marks)

Symbol	**Meaning / argument order**
Learner(x)	x is a learner
Course(c)	c is a course
Enrolled(x,c)	learner x is enrolled in course c
Completed(x,c)	learner x completed course c
Prerequisite(p,c)	course p is a prerequisite for course c
NLP, AI, Logic	Example course constants
x, c, p	Learner, course and prerequisite variables

2.2 Formalisation (3 Marks)
(i) Every learner enrolled in NLP has completed at least one prerequisite:

$\forall x\ [(Learner(x) \wedge Enrolled(x,NLP)) \rightarrow \exists p\ (Course(p) \wedge Prerequisite(p,NLP) \wedge Completed(x,p))]$

(ii) Some learner completed every course offered by the department:

$\exists x\ [Learner(x) \wedge \forall c\ ((Course(c) \wedge OfferedByDepartment(c)) \rightarrow Completed(x,c))]$

Here OfferedByDepartment(c) is an additional predicate needed to express "offered by the department".

(iii) No learner completed a course that they were not enrolled in:

$\forall x \forall c\ [(Learner(x) \wedge Course(c) \wedge Completed(x,c)) \rightarrow Enrolled(x,c)]$

2.3 Quantifier-Scope Challenge (2 Marks)
Original: $\forall x\ (Learner(x) \rightarrow \exists c\ (Course(c) \wedge Enrolled(x,c)))$. Every learner has at least one course, and the course may differ by learner.

Modified: ∃c (Course(c) ∧ ∀x (Learner(x) → Enrolled(x,c))). There is one course in which every learner is enrolled.

Finite model: Learners = {Asha, Bala}; Courses = {NLP, AI}; Enrolled = {(Asha,NLP), (Bala,AI)}. The original is true, while the modified formula is false. Therefore, changing the quantifier order changes the meaning and the formulae are not equivalent.

2.4 Consistency Test (2 Marks)

Knowledge base facts: Learner(Asha), Learner(Bala), Course(NLP), Enrolled(Asha,NLP), Completed(Asha,NLP). Rule: for every x and c, if Learner(x), Course(c), Enrolled(x,c) and Completed(x,c), then PassedOrCompleted(x,c) (here completion is treated as the stored completion result).

Query: Completed(Asha,NLP). Result: TRUE because the exact fact is present in the knowledge base. Query: Completed(Bala,NLP). Result: NOT PROVED because no matching fact is stored. This is not proof that Bala failed; it only means the knowledge base has insufficient evidence. The inference is traceable to the stored facts and stated rule.

Task 3 – Linguistic Concepts and Syntax-Driven Semantic Analysis (9 Marks)

Sentence: "The analyst noticed the student with the camera near the library."

3.1 Linguistically Relevant Concepts (2 Marks)

Concept	Example	Semantic function
Entities	analyst, student, camera, library	Participants and locations
Predicate	noticed	Perception event
Modifier	the	Determiner; identifies noun phrase
Prepositional relations	with the camera; near the library	Possible possession/instrument and location relations
Semantic roles	analyst=experiencer; student=theme	Connects participants to noticed event
Attachment points	student NP, noticed VP, student/event location	Determines which entity or event a phrase modifies

3.2 Parse-to-Meaning Procedure (3 Marks)

Parse A – both phrases modify the student: "the student [with the camera] [near the library]".

Trace: analyst → NOTICED event; student → theme; with-camera → HAS(student,camera); near-library → LOCATION(student,library).

Representation: NOTICE(e1) ∧ EXPERIENCER(e1,analyst) ∧ THEME(e1,student) ∧ HAS(student,camera) ∧ NEAR(student,library).

Parse B – “with the camera” modifies the noticing event and “near the library” locates the event.

Trace: analyst → NOTICED event; student → theme; with-camera → INSTRUMENT(e1,camera); near-library → LOCATION(e1,library).

Representation: NOTICE(e1) ∧ EXPERIENCER(e1,analyst) ∧ THEME(e1,student) ∧ INSTRUMENT(e1,camera) ∧ LOCATION(e1,library).

The readings diverge when the prepositional phrases are attached: HAS(student,camera) and LOCATION(student,library) in Parse A, versus INSTRUMENT(e1,camera) and LOCATION(e1,library) in Parse B.

3.3 Composition and Type/Role Checks (2 Marks)

Word meanings are first assigned to lexical entries. The noun phrases form analyst, student, camera and library entities. The verb “noticed” creates a perception event with an experiencer and a theme. Prepositional phrases then add relations to either a noun phrase or the event.

- Type check: an INSTRUMENT relation must connect an event to an object that can plausibly function as an instrument.
- Role check: NOTICE should have an experiencer and a perceived theme; a location phrase must attach to an entity or event that can be located.

3.4 Error Challenge (2 Marks)

A syntactically possible but implausible reading is that the camera is the person who noticed the student. The parse may be structurally possible if the system confuses attachment or roles, but the semantic type of camera does not normally satisfy the experiencer role. The system should flag this mismatch. Refinement: use selectional preferences (a noticing experiencer is typically animate) and discourse context. The system should retain alternatives if the evidence is insufficient.

Task 4 – Lexemes, Senses and Word Sense Disambiguation (8 Marks)

4.1 Lexeme and Sense Inventory (2 Marks)

The lexeme LIGHT groups word forms such as “light”. The word form is the observed token; the sense is the meaning selected in context.

Example	Part of speech	Sense label	Meaning
Turn on the light.	Noun	LIGHT_ILLUMINATION	A lamp or source of illumination
This bag is light.	Adjective	LIGHT_LOW_WEIGHT	Not heavy
The light meal was served early.	Adjective	LIGHT_SMALL_MEAL	Not heavy or substantial in quantity

4.2 WSD Design (2 Marks)

Input: sentence, target token and its part of speech. Candidate senses: illumination, low-weight and small/not-heavy meal. Context features include nearby nouns and verbs, adjective position, and words such as switch, lamp, bag, carry, meal, food and served. The rule-based decision assigns a score to each sense using matching context cues. It returns the highest unique positive score; otherwise it returns UNCERTAIN. Output format: target, POS, selected sense, cue evidence and confidence/score.

4.3 Apply and Challenge (2 Marks)

Sentence	Evidence	Output
Turn on the light.	turn on; action commonly used with a light source	LIGHT_ILLUMINATION
This bag is light.	bag; adjective describes weight	LIGHT_LOW_WEIGHT
The light meal was served early.	meal; served	LIGHT_SMALL_MEAL
The suitcase is light enough to carry.	suitcase; carry	LIGHT_LOW_WEIGHT

Failure case: “The light was unusual.” The context does not clearly show whether light means illumination or a particular visual quality. The method should return UNCERTAIN or request more context instead of forcing a sense.

4.4 Evaluation (2 Marks)

Labelled test set (8 instances):

ID	Sentence	Gold label
1	Turn on the light.	LIGHT_ILLUMINATION
2	The lamp gives bright light.	LIGHT_ILLUMINATION
3	This bag is light.	LIGHT_LOW_WEIGHT
4	The suitcase feels light.	LIGHT_LOW_WEIGHT
5	The light meal was served.	LIGHT_SMALL_MEAL
6	We ordered a light breakfast.	LIGHT_SMALL_MEAL
7	The light was unusual.	UNCERTAIN
8	The box is light to carry.	LIGHT_LOW_WEIGHT

Annotation policy: label the intended sense using the full sentence; use UNCERTAIN if the sentence does not provide enough evidence. For an example result of 7 correct out of 8, accuracy = 7/8 × 100 = 87.5%. Class imbalance can make accuracy misleading because a system may

perform well on the common sense and poorly on rare senses. Ambiguous annotation can also make a correct alternative reading appear incorrect. Report per-sense performance and inspect disagreements.

Task 5 – Semantic Information Retrieval (8 Marks)

5.1 Query and Document Semantics (2 Marks)

Query: “How can a system identify the intended meaning of a word in context?” Intent: find methods that select the correct sense of an ambiguous word by using surrounding linguistic context.

ID	Candidate document summary	Label
D1	Word sense disambiguation uses context to identify the intended meaning of an ambiguous word.	Relevant
D2	A dictionary lists definitions and example usages of words.	Distractor / partial
D3	Contextual language models select word meanings using surrounding words and sentence structure.	Relevant
D4	A bank provides savings accounts, loans and cash services.	Distractor
D5	Lexical semantics studies word meanings, senses and relations between concepts.	Relevant

The query can be represented with concepts WORD, INTENDED_MEANING, CONTEXT, SENSE and DISAMBIGUATION. Documents are represented with their terms, concept labels and topic summary. The relevant documents may use “word sense disambiguation” or “contextual language models” instead of repeating the query exactly.

5.2 Ranking Strategy (2 Marks)

Use lexical overlap and semantic concept similarity. Each value is normalized from 0 to 1.

Score(d) = 0.40 × LexicalOverlap(d) + 0.60 × SemanticSimilarity(d).

Semantic similarity receives the higher weight because the query asks for a meaning-based method. If two documents tie, prefer the one covering more query concepts; if still tied, sort by document ID for reproducibility.

5.3 Retrieval Trace (2 Marks)
The supplied Python demonstration calculates lexical overlap and a small concept-cue score for six sample documents. Its actual score uses 0.40 × lexical + 0.60 × semantic. The values are transparent teaching scores, not scores from a trained embedding model.

Document	What it tests
D1	Direct WSD explanation; expected to rank highly.
D3	Contextual model explanation; meaning match even with different wording.
D5	Related lexical-semantics concepts; useful but broader.
D2	Dictionary definitions share word/meaning terms but may not explain contextual selection.
D4	Contains “bank” and “cash”, but does not answer the WSD query.
D6	Spelling correction is an NLP task but not word-sense selection.

A semantically relevant document can outrank a keyword distractor because the semantic signal rewards coverage of the intended concepts, not merely repeated words.

5.4 Evaluate (2 Marks)
At a cutoff of 4, suppose the system retrieves D1, D3, D5 and D2. D1, D3 and D5 are relevant; D2 is not counted as relevant because it lists definitions but does not explain contextual disambiguation. If there are 3 relevant documents in the collection, precision@4 = 3/4 = 0.75 and recall@4 = 3/3 = 1.00.

False positive: D2, because it discusses definitions but not how context selects a sense. False negative: a relevant document using “polysemy resolution” may rank below the cutoff due to low literal overlap. Improve the system by adding task-intent features and expanding query concepts with synonyms such as polysemy and WSD.

Task 6 – Integrated Semantic Analysis Design Challenge (8 Marks)
Sentence: “I watched the crane by the bank, then it moved.”

6.1 Model the Ambiguity (1 Mark)
“Crane” may mean a bird or a lifting machine. “Bank” may mean a river edge or a financial institution. “It” may refer to the crane, the bank or another entity. Context such as construction activity, water, wildlife, or a bank building is needed.

6.2 Candidate Meaning Structures (2 Marks)

Candidate A – bird and river edge: the speaker watched a crane bird near a river bank; the bird then moved.

Representation: WATCH(e1) ∧ AGENT(e1,speaker) ∧ THEME(e1,crane_bird) ∧ NEAR(crane_bird,river_bank) ∧ MOVED(crane_bird).

Candidate B – lifting machine and financial-bank area: the speaker watched a construction crane near a bank building; the machine then moved.

Representation: WATCH(e1) ∧ AGENT(e1,speaker) ∧ THEME(e1,crane_machine) ∧ NEAR(crane_machine,financial_bank_building) ∧ MOVED(crane_machine).

6.3 Syntax and WSD Trace (2 Marks)

The parser identifies "the crane" as the object of watched and "by the bank" as a location phrase. The pronoun "it" is resolved by checking agreement and discourse salience; both crane readings are singular, so grammar alone cannot decide. Context words such as wetland, feathers or river support the bird reading; construction, lifting or operator support the machine reading. If evidence is insufficient, preserve both candidates or ask a clarification question.

6.4 Semantic Retrieval (2 Marks)

For Candidate A, form a query with CRANE_BIRD, RIVER_EDGE, WILDLIFE and HABITAT. Rank these documents:

Rank	Document	Reason
1	Crane birds in river wetlands	Matches bird sense, river location and wildlife topic
2	Wetland habitat and water birds	Strong context match, although it does not focus only on cranes
3	Construction crane lifting safety	Lexical match for crane but wrong sense
4	Bank accounts and financial services	Lexical match for bank but wrong sense

The ranking uses sense compatibility and concept coverage. A document is not selected merely because it repeats "crane" or "bank"; its sense must match the chosen interpretation.

6.5 Adversarial Refinement (1 Mark)

Misleading context: "The crane was beside the river during a construction project." River suggests a bird, while construction suggests a machine. A simple cue-count system may tie or choose incorrectly. The design should keep both readings, use broader discourse context and ask for clarification if confidence remains low. This reduces forced, unsupported decisions.

Conclusion

The two parts show how NLP systems move from sentence meaning to structured representations, logical reasoning, syntax-based interpretation, word-sense disambiguation and semantic retrieval. Representations make roles and relations explicit, while quantifiers express general and specific claims. Ambiguous phrases require alternative parses and context-based decisions. Retrieval should consider meaning as well as exact words, and its performance must be measured using suitable test data, precision, recall and accuracy. A reliable semantic system should preserve uncertainty, explain its decisions and be tested with misleading examples.

Appendix – Python Project and Execution

The optional ZIP project contains main.py, README.md, requirements.txt, sample_output.txt and a screenshot of the program output. It uses only the Python standard library. It demonstrates the main Part II ideas with transparent rules and small sample data.

Run command: python main.py

Limitation: the code is a compact educational prototype. It does not include a trained parser, neural language model, production-grade ontology or live search engine. The written report contains the complete reasoning required by the assignment.